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Bitcoin Is Venice

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Rhapsody on a Theme of Nakamoto



n.b. this essay has since been adapted into a standalone chapter in the book, Bitcoin Is Venice (Amazon link here) by Allen Farrington and Sacha Meyers.

NOT FINANCIAL ADVICE. DO YOUR OWN RESEARCH.

Quentin Skinner's monumental overview of the development of early modern political philosophy, *The Foundations of Modern Political Thought*, begins with the following lines,

"*As early as the middle of the twelfth century the German historian Otto of Freising recognised that a new and remarkable form of social and political organisation had arisen in Northern Italy. One peculiarity he noted was that Italian society had apparently ceased to be feudal in character.*"

While Skinner's concern is political philosophy and not economic history, it is easy enough to identify that these social changes were made possible by a nascent form of capitalism. As the great medievalist Henri Pirenne commented on the period and region in his *Medieval Cities*,

"*Lombardy, where from Venice on the east and Pisa and Genoa on the west all the commercial movements of the Mediterranean flowed and were blended into one, flourished with an extraordinary exuberance. On the wonderful plain cities bloomed with the same vigor as the harvests. The fertility of the soil made possible for them an unlimited expansion, and at the same time the ease of obtaining markets favored both the importation of raw materials and the exportation of manufactured products. There, commerce gave rise to industry, and as it developed, Bergamo, Cremona, Lodi, Verona, and all the old towns, all the old Roman municipia, took on new life, far more vigorous than that which had animated them in antiquity*."

Pirenne added that the rise of these cities, which was predicated on commercial and industrial expansion,

"*strongly stimulated social progress. It made no less a contribution in spreading throughout the world a new conception of labor. Before this it had been serf; now it became free, and the consequences of this fact, to which we shall return, were incalculable. Let it be added, finally, that the economic revival of which the twelfth century saw the flowering revealed the power of capital, and enough will have been said to show that possibly no period in all history had a more profound effect upon humanity*."

And wouldn't you know it, but feudalism seems to be making a comeback. Joel Kotkin introduces his pithy tract, *The Coming of Neo-Feudalism*, anticipating this re-emergence:

"Of course it will look different this time around: we won't see knights in shining armor, or vassals doing homage to their lords, or a powerful Catholic Church enforcing the reigning orthodoxy. What we are seeing is a new form of aristocracy developing in the United States and beyond, as wealth in our postindustrial economy tends to be ever more concentrated in fewer hands. Societies are becoming more stratified, with decreasing chances of upward mobility for most of the population. A class of thought leaders and opinion makers, which I call the "clerisy," provide intellectual support for the emerging hierarchy. As avenues for upward mobility are diminishing, the model of liberal capitalism is losing appeal around the globe, and new doctrines are arising in its place, including ones that lend support to a kind of neo-feudalism."

Not all, but certainly *some,* of these afflictions can readily be attributed to the normalized strip mining of capital in the pursuit of ever more leveraged "growth" I outlined in Part Two of this series, *The Capital Strip Mine*. Those

who do not own hard assets are increasingly tending to drown in debt from which they will realistically never escape, unable to save except by speculation, and unable to afford the inflation in the essential costs of living that does not officially exist. What amounts to an "official" message is the likes of Christine Lagarde (then president of the IMF and now of the ECB) musing that, "*we should be happier to have a job than to have our savings protected,*" and the World Economic Forum projecting that, by 2030, "*you will own nothing, but you will be happy.*" You will *use things* that *somebody owns,* mind you. But that somebody will not be you.

If we were to believe that these people actually mean what they say, and that the strip mining of capital is not going to stop — indeed, that it *cannot stop* — we might be as similarly inclined as Otto of Freising to look for any sprouts of civilization that manage to advance beyond our rebooted feudalism. There may end up being a variety of reasons that different groups avoid this state. I think that, for some, the reason will be Bitcoin.

What does that mean? I am sure it seems hyperbolic to most, if not outright ludicrous, but it's actually quite prosaic. It means that those social units that voluntarily choose to liquidate their positions in self-referentially mispriced toxic loans in favor of a global, digital, sound, open-source, programmable money will be in a position to accumulate long-term oriented capital at a disproportionate rate to those who do not. They will have a superior economic foundation from which to build healthy social and political institutions, which will contrast to those left behind as medieval Venice did to the remnants of the Western Empire. This could be true at any and every scale. It could be an individual, a family, a friend group, a neighborhood, a company, a city, an industry, a country, or the entire world. We will have to wait and see.

Of course, it could be nobody. It could fail altogether. I say this primarily to guard against accusations of blind faith, speculative mania, and fundamental unseriousness. But I *don't say it* to feign intellectual sophistication with post-hoc unfalsifiable fence-sitting. As if this wasn't entirely clear already, I am very happy indeed to be on the record as saying it is overwhelmingly likely Bitcoin will succeed. And so, while there are good reasons it might fail, "*it's dumb*," and, "*I don't like it*" are not among them. In order to sensibly articulate the reasons why it might fail you have to understand it in the first place. Most do not. As I explained in Part One of this series, *Wittgenstein's Money*, most do not even know what it is they are looking at. Nor are they likely to any time soon because they don't want to see it. As philosopher of science Norwood Russell Hanson might say, their perception is *theory-laden*. Also, their theories are wrong. Oops.

And so, in the spirit of such colorful outrage as "*it's a Ponzi scheme!*", "*it's a waste of energy!*", and, "*it's backed by nothing!*", I will round off this trilogy with my own set of colorfully outrageous metaphors to try to help people understand *what is actually happening right now*, and why things seem exactly like they should.

So outrageous they might just be accurate …

Bitcoin is Ariadne



> *Anyone who accumulated large amounts of wealth while remaining independent of military-political command structures faced the problem of safeguarding what he had gained. Unless a merchant could count on the protection of some formidable man of power, there was nothing to restrain local potentates from seizing his property any time his goods came within reach. To gain effective protection was likely to be costly — so costly as to inhibit large-scale accumulation of private capital.*
>
> *- William McNeill*

Bitcoin is often framed as "competing" with fiat currency. This is true in a sense but I fear there is a rhetorical danger of invoking the wrong kind of "competition". It is not a *fight,* for example. There is no conflict. Bitcoin is not trying to damage or sabotage its opponents, because it isn't *trying* anything and it knows no opponents. It has no awareness whatsoever of who might oppose it or why. It is simply an alternative; an exit valve; an opt-out. It is competing only insofar as it is proving to be a far superior alternative. It is not a sword for Theseus to fight the Minotaur, but a thread to follow to exit the labyrinth. Bitcoin is Ariadne.

There will be tremendous value in normalizing this rhetoric amidst the likely growing chorus of opposition desperate to smear Bitcoin as inherently nefarious, or hostile, even. Opponents must be forced to explain what is wrong with people interacting freely, and why true goodness can only follow from coercion, in their understanding. Should those who have found a way out of the unbearable labyrinth of capital strip mining not take it? What do they owe the Minotaur?

Does anybody really believe that, having fully understood the choice they face, any individual would choose to save in a self-referentially mispriced toxic loan rather than a provably sound digital bearer asset? Or, more simply still, that they will think it makes less sense to hold money that is a pure asset than money that is literally *defined as* a liability? Why not opt into a financial system that is built on trustless verifiability rather than unverifiable trust?

Threats of violence, perhaps? After all, the only way to "seize" properly secured bitcoin is through torture. In, *The Pursuit of Power,* historian William McNeill interprets efforts in early modern Europe to industrialize and standardize weaponry and military drill as having the effect that, "*the*

magnitude and controllability of organized violence per tax dollar went up — spectacularly." It seems reasonable to suggest the potential for a similarly spectacular *fall* in such returns of late, as German prosecutors and Reuters alike recently discovered:

or maybe they didn't discover it, but tweeted this anyway. who knows ...

It is worth working through the optics of any decision to engage with Bitcoin in a truly hostile manner, because it is certainly coming. McNeill reminds us that, even some seven-hundred-or-so years ago, "*the breakdown of established patterns of conduct always appears deplorable to a majority of those who witness it.*" By no means do I have a utopian outlook on this subject — rather, it is something of an intellectual rite of passage to accept the nonzero utility of dystopian paranoia. Bitcoin will be banned, many times, in many places. But a ban is an open admission of practical and moral failure and is arguably the best advertisement of all. A ban is the Berlin Wall; fragments of any ban will one day become souvenirs of the folly and cruelty of repression. Bitcoin

doesn't force anybody to stay. They come, and then they stay, because they want to — because it is both practically and morally superior.

As with East and West Berlin, it is also worth working through the likely ripples on society at large of the foundational difference of valuing and enshrining voluntary interaction. Yes, Bitcoin has different mechanics — I will get to this further down — but from these mechanics follow different behaviors; from these behaviors, different cultures; from these cultures … who knows?

I don't profess to know, but I can offer some ideas. First, we are wholly unprepared for the societal implications of making most wealth and much capital *entirely* mobile. We have been edging in this direction for decades as *software has eaten the world*, as Marc Andreessen, professional baller, famously put it. As I articulated here, Andreessen's text remains probably the most important treatise on finance written this century, and yet many in finance have not read it, and many who have think it is not about finance but about tech. It is only about tech insofar as by "tech" you mean "software", by "software" you mean "everything", and by "everything" you mean "finance". So, you are right, but in at least three different wrong ways. Maybe more.

My preferred philosophical abstraction of Andreessen's argument would be something like the following: software is productive capital for which the raw ingredients are coherent human thoughts. This has recreated the independent skilled-laborer-cum-entrepreneur as a class of economic agent whose capacity for capital creation is human, not financial. This class has arguably been minimized in the economic landscape at large since the Industrial Revolution morphed the Venetian blueprint of capitalism to its vastly more complex successor stage dependent on organizing and directing labor around immobile capital. Such agents have vast bargaining power over

financial capitalists, which they tend to exercise currently by demanding equity. But note, the equity stake grounds capital creation in the existing financial system. This power was only ever forward-looking — such workers could bargain over and make mobile wealth they were yet to create.

But Bitcoin has severed the final link. Vastly more capital is, in theory, now mobile as it no longer needs to be moored to a given financial system. By "everything" we need not mean "finance", but can actually mean "everything", particularly since a silver lining of the nightmare of lockdowns seems to have been normalizing knowledge work from just about wherever the workers want rather than a handful of unliveable metropoles. Kotkin laments that,

"*Rather than a base for upward mobility, the great cities have largely become magnets for those who are already well-to-do. Few working-class or middle-class families can now afford to move to places like Paris, London, Tokyo, New York, San Francisco. Many former residents, like Chicago's black middle class, have left to make their future elsewhere. Many who still work in those cities are forced into intolerably long commutes. As the middle class dwindles, it leaves behind a marginal urban population who depend on the city for a livelihood but often can barely get by.*"

But likely no longer. And of course, wherever this high-skilled, newly mobile capital-cum-labor recongregates, all other forms of work will be viable as well — this need not be understood at surface level as an elitist prediction, but rather as baby steps towards feasible localism, at long last.

Physical capital still matters, clearly. So does cultural capital. These are so obvious as to be weird to need to point out. But those in a position to extract protection rents on physical capital, likely with the allure of cultural capital,

will need to adjust to this new reality. Sticks are out, carrots are in. What are you gonna do about it? Build a wall? Good luck with that.

In the remarkable essay, *Economic Consequences of Organized Violence*, historian Frederic Lane emphasizes the importance of sovereign competition in using and controlling violence in an era of more mobile capital than we are used to today:

"If all the tribute was used for conspicuous consumption, a term which seems particularly appropriate for the court of a prince of the ancièn regime*, growth was slowed by lack of investment. Merchants who gained protection rents from international trade and colonization, although not entirely inconspicuous in their consumption, probably had a lower propensity to consume. If so lower profits for governments and higher profits for trading enterprises meant more capital accumulation and more growth."*

McNeill similarly observes that in the wake of the eleventh-century upsurge of Venetian and Genoese private commercial activity in the Mediterranean,

"Rulers of old-fashioned command societies were simply unable to dominate behavior as thoroughly as in earlier times. Peddlers and merchants made themselves useful to rulers and subjects alike and could now safeguard themselves against taxation and robbery by finding refuge in one or another port of call along the caravan route and seaways, where local rulers had learned not to overtax the trade upon which their income and power had come to depend."

We may well be heading back to such a dynamic, with the wilds of the Internet the spiritual successor to the high seas.

But what does finance look like in such a society? It certainly doesn't follow from the above that finance *disappears*. Surely it just changes - but to what? I think there are two helpful strands of answers. The first is *programmable*, which, by its nature is unpredictable except by its potential. Analogies to the early web are cliché, but with perfectly good reason. With open access and a programmable interface, who knows what will be invented? Who knows how quickly inventions will be iterated and combined? The second is *Islamic*.

Bitcoin is Halal

photo by **Abdullah Faraz**, via Unsplash

> *"An individual can be arrested for 'manufacturing' money in his own home but the commercial banking system is given the full protection of law in doing what amounts to the same thing. There is no justice in this … There are those who say that we must develop an Islamic alternative to modern commercial banking. But why* must *we do so? The Islamic alternative to the cigarette industry is no cigarette industry, and were we to remain true to our principles we might realize that the Islamic alternative to commercial banking is no commercial banking."*
>
> *- Tarek El Diwany*

That Bitcoin can potentially be considered concordant with the teachings of Islamic finance is an insight I owe to conversation with Saifedean Ammous.

What we mean by this is roughly as follows: as Bitcoin is a digital bearer asset and not a debt instrument, its natural state of safe custody is outside financial institutions. Also, without the ability to mint new Bitcoin as and when politically convenient, deposit insurance is impossible and loan origination requires the prior provision of liquid capital. Hence prospective intermediaries would not be able to guarantee investor protection from loss arising from the debtor's activities.

In combination, it becomes comparatively much less likely that contributors of capital will accept a fixed upside and unlimited downside, particularly given the near-certainty of deflation rather than the contemporary norm of inflation that must be forcefully chased with low-volatility capital appreciation. Would-be depositors will either save unilaterally or demand the shared upside of equity, for the most part. Supply will contract and swathes of demand for leverage will be priced out of whatever market is left.

What remains of interpersonal, non-programmatic banking is likely to look very similar to the proscriptions of Islamic finance, with a dramatically reduced role for debt, and a focus on *risk-sharing* rather than *risk-transfer*. Bitcoin is halal.

To be clear, the reasoning that arrives at this outcome is quite different. In Islam, interest on monetary debt (riba, الربا) is unlawful (haram, حرام) on ethical grounds whereas I suggest it is unlikely to emerge to any great extent in a Bitcoin standard on purely economic grounds given the risks will (finally!) be properly priced, hence can be legitimately shared rather than dishonestly transferred. Even still, the resulting behavior has a clear ethical resemblance to the cultural norms in the Bitcoin community, centered first and foremost on low time preference. Consider the Islamist scholar Abul Alā Mawdūdī's (ابو الاعلی مودودی) exhortation that,

"*it is incumbent on every member of the Muslim community to live within his means. He is forbidden to let his expenditure exceed his income, thus compelling him to stretch his hands out to others in order to sustain his extravagance, use unfair means to grab the wealth of others or become indebted to others to help finance his unending needs and, by consuming his resources in clearing his debt, eventually join the ranks of the destitute.*"

And even if we contest the underlying ethical objection, more similarities in attitude at a higher level are readily forthcoming. We might think, for example, that even in a Bitcoin standard there will still be *some* willing supply and demand for interest-bearing, at-risk capital; why then should consensual exchange be prevented? The Islamic response hinges on the *haram* feature of what is deemed "speculation"; given the inherent uncertainty of entrepreneurship, the commitment to pay interest on monetary debt finance is unavoidably fraudulent given the debtor does not *and cannot* really know she can meet her obligations. Only *pari passu* equity stakes are fair and honest, and furthermore, introduce no disharmony of incentives or outlooks between the capital providers. Once more, we may differ in our understanding of the ethics of such contracts, but Mawdūdī's reckoning of the consequences of the proliferation of such "speculation" has a clear appeal:

"*Due to the absence of a reasonable and healthy relationship of participatory cooperation between the capitalist* [creditor] *and the entrepreneur* [debtor]*, the global economy suffers tremendously and faces alternative highs and lows that adversely affect the world's economic health. The capitalist's stranglehold had helped to boost the spirit of speculation and minting of money through interest. this has naturally poisoned the bilateral relationship between capital and enterprise, and the raising and lowering of interest rates are now done in such a way as to keep the entire world's economic health always in risk.*"

By simply substituting out "all debt" for "money as debt" and similarly tracing out the consequences, Bitcoiners would likely agree entirely. Economist and noted Islamic Finance scholar Mohammad Siddiqi dryly makes this connection in, *A Vision for the Future of Islamic Economics*, noting that, *"almost all money in circulation is interest bearing debt transferring wealth from fund users to the owners of capital. It is not technically necessary for society's means of payment to play this role."* In, *The Problem with Interest,* Tarek El Diwany makes a strikingly similar point to my own in *The Capital Strip Mine*, all the more remarkably from an entirely different basis, that, "*polluted rivers, festering rubbish tips and resource-depleted seas may be just the first installment of the price that is paid for entering in to a race with compound interest*."

To a limited extent, believe it or not, the IMF seems to agree as well. In a 2010 working paper titled, *The Effects of the Global Crisis on Islamic and Conventional Banking: A Comparative Study*, Maher Hasan and Jemma Dridi conclude that Islamic banks' asset-based, rather than debt-based, operations, "*make their activities more closely related to the real economy and tend to reduce their contribution to excesses and bubbles.*" The paper also includes the unintentionally hilarious deadpan explanation that,

"*the profit/loss-sharing nature of investment deposits provides Islamic banks with an additional buffer. However, this feature was not tested in the crisis given that most banks remained profitable. In addition, in the context of the crisis and given the loose monetary stance in most countries, this feature is likely to put Islamic banks' profitability at a disadvantage compared to conventional banks.*"

The closest analogs to "loans" as we might think of them are qard al-hasan (قرض الحسن), a straightforward interest-free, benevolent loan, and sukuk (صكوك), a kind of pooled, fixed-term equity investment — curiously more like

the Venetian *colleganza* than any common contemporary instrument. Collateral is uncommon in either case, but where it is taken there is a final intriguing comparison to be made: the collateral must be transferred to the possession of the creditor for the term, hence in the case of default, no repossession occurs. As with everything in Islamic finance, the ultimate rationale is simply fairness and justice; those who default on a mortgage are not kicked out of their home because the home is in possession and use by the debtor and could not have been *halal* collateral to begin with. The connection to Bitcoin is the likely incentive in a healthily deflationary environment not to recourse to hard assets to store value over the long-term, combined with an increased background incentive in financial transactions for all parties to access and retain sound money:

One need not personally subscribe to all, or any, tenets of Islamic finance to appreciate a broader point than its admittedly tenuous connections to Bitcoin. The study of Islamic economics and finance is intriguing as it is, to my mind, the only systematized, contemporary, and *successful* alternative to what we might call "Western" financial precepts, now, of course, near-globalized and seemingly omnipresent in commerce. Note, for example, the IMF describes "Islamic Banks" as opposed to "Conventional Banks" (and I stress "successful" above, by the way, to properly distinguish and contrast Islamic economics to socialism, of which, the reader might be interested to know, Mawdūdī's *Mankind's Economic Problems and their Islamic Solutions* provides possibly the most concise and effective refutation I have ever read, fusing Mises and Havel in a two-page polemic).

A common difficulty with the conceptual challenge posed by Bitcoin is that this omnipresence makes it difficult to think rigorously in terms outside the framework of mainstream Western finance at all. Western finance *is water*. But perspective *can be* achieved, and precepts *can be* challenged. Mawdūdī,

Siddiqi, and El Diwany challenged them, Nakamoto challenged them, and the reader owes it to herself to challenge them too. Elevating morality above perceived efficiency makes for a profound starting point.

Bitcoin is Gravity



picture by **Wikilmages**, via Pixabay

> *"An amount of money lent to a government, and the interest amount charged, is assumed to be risk-free because it is in turn assumed that a government can tax, borrow, or print further amounts of money to pay its debt. These three options are indeed available to a modern government, but one must not ignore the fact that the government has no access to risk-free rates of return when investing the borrowed money. The above mentioned options are in fact nothing more than means of passing on the bill to others when the fact of a non-risk-free physical system eventually reasserts itself."*

- Tarek El Diwany

To date, it has been very difficult to conceptualize what value, exactly, has peacefully opted out of fiat and into Bitcoin. Pricing only happens at the margin, and marginal fiat exchanged for bitcoin is just a bank liability that the bank relabels. Bitcoin exchanged at the margin likely engenders lower time preference, as discussed above, hence lower consumption of garbage, but which is conceivable only counterfactually.

This will change when people start selling not just their fiat, and not just their time, but when they start liquidating real assets. Gold will probably be the first victim, for readily understandable reasons as Bitcoin is an upgrade in almost every respect. But gold is not systemically important. This shift will be noticeable but not otherwise impactful. When the accumulation drive hits short-term credit, real estate, and passive equity, that is when the party will really start.

These three are artificially large asset classes given they are de jure productive, hence cash generative and priced on yield, yet de facto speculative savings instruments given long-term saving with fiat is impossible. But more vitally still, they *are* systemically important. Their prices, in aggregate, affect capital formation. The TLDR of Part Two, *The Capital Strip Mine*, is that these prices are wrong, and hence the capital is being strip-mined as quickly as it is being formed. Reversing this is the long-term hope, but anticipating the short-term mechanics of this reversal is another matter entirely.

The key insight is that if these assets were *actually* priced on yield, the kind of flows I anticipate would have no impact on long-term holders beyond minor disappointment. But because they are not, any substantial outflow can

easily become a self-fulfilling prophecy. Corporations use short-term credit as ersatz cash with an inflation-hedging yield, however minuscule these days. But this is not an "investment". There is no upside, but just-sure-enough downside protection. If the downside protection disappears then the entire proposition evaporates — and note this could easily happen without substantial selling but simply a neglect to continue buying, given the whole point of short-term credit is that it continuously rolls over. What would likely happen next is central banks stepping in to "support" these markets with asset purchases, which, of course, is the best imaginable endorsement of Bitcoin's utility.

Behind all of this is the seemingly straightforward question of bitcoin's "fair value". There will always be a hesitancy to shift savings from something as well understood and naturally priced as short-term credit to bitcoin on the basis of being entirely unsure of how to compare bitcoin's price to its "fundamentals". What will gradually be realized is that, with Bitcoin, the traditional relationship is inverted. I do not think it is quite accurate to say that bitcoin's price *is* its fundamentals, but certainly its price is a largely reflexive function of its fundamentals: as the price goes up, the fundamentals go up (and we must be mindful also that as the price goes down, the fundamentals go down. Sustained attack that drives the price down for long enough is by far the biggest risk). Bitcoin was weakest when smallest, but less so the more time passes. Bitcoin is a black hole sucking unsustainably artificial value beyond its event horizon. As it grows, so does its pull. Bitcoin is gravity.

The "store of value realization" argument for its gravitational pull is by far the most obvious, the least creative, and is only scratching the surface of its likely continued evolution. Consider the implications of deepening liquidity, which, note, is subtly different from "price" alone. This is a necessary

precondition for increasingly large purchases in the first place. MicroStrategy could not have done what it did a year earlier. Apple and Berkshire (dare I say?) still cannot do what they likely one day will.

But market deepening has far more interesting implications. It enables Strike, for example, Zap's soon-to-be-widely-copied, soon-to-be annihilation of FX markets:

Strike combines ever-tightening spreads on fiat liquidity pools with Lightning's instant settlement and relative programmability to offer unmatchable FX transfers. There are several astonishing features here that it is worth making absolutely sure we understand.

First, this service cannot be matched within the fiat settlement infrastructure. I don't mean that it is difficult; I mean that it is impossible. Interbank payments with the same currency and within the same

jurisdiction can be more or less free and instant, and in many places are, since all this amounts to is relabelling a bank liability or, at worst, a net flow between mutual counterparty banks that can be batched and properly settled at enormous scale, hence offered to the end-users at low or zero cost. But across currencies, jurisdictions, or both, this is impossible — fundamentally *because* fiat is a debt instrument. What we might think of as a simple "payment" in this context is really more like a credit relay. Each party needs to trust the next party in the chain, and price not only the operational expense but this perceived risk, before passing it on, given the actual claim will be settled much, much later. And payment *streaming*? Fuhgeddaboudit. Not in your wildest dreams. With Strike, none of this is relevant. Lightning has no lower bound on value and settles instantly, and that's the end of that.

This service does not expose the user to the price of bitcoin *at all*. And yet, the fact of its existence and usage deepens the markets, which directly contributes to bitcoin's fundamentals, hence price, increasing. And if heavy users of this system one day decide they'd prefer to keep the transfers they receive in sound, open-source, programmable money, well, that makes the process even simpler still …

It doesn't even end there. The Lightning infrastructure is still young and small and it needs staked value to grow. What better way to put bitcoin-denominated capital to work than seeking a return on competitive liquidity and routing? As bitcoin's fiat value grows, so do the incentives to contribute to scaling Lightning, which increases the efficiency of fiat payments routed via Strike-and-others over Lightning, which increases the depth of the fiat markets for bitcoin. And the more Lightning scales, the more the prospect of payment streaming opens up opportunities for better funding decentralized infrastructure — for example, incentivizing running Tor exit nodes, the storage and routing building blocks sought by the likes of Sci-Hub, accessible

and portable economies in gaming, off-platform content monetization and advertising-free content-driven apps, but also things literally nobody has yet imagined. All of which increases Lightning's utility, which increases Bitcoin's utility. The more falls into this orbit, the bigger the orbit gets. The Jevons paradox in the metaverse!

Deeper markets also indirectly legitimize lending fiat against bitcoin reserves. While nominally tailored to allowing synthetic institutional leverage, normalizing this service will reduce the incentive for anybody to ever sell, in inverse proportion to how widely accepted bitcoin is for regular payments at a given time. It will be Pierre Rochard's speculative attack, but without even requiring awareness or intention. It will just be the sensible thing to do. If or when miners are able to access this service to pay for electricity, even partially, marginal supply will evaporate. It will also make increasingly viable credit card rewards programs, salary allocations, and, once again, things literally nobody has yet imagined, which might read as negligible in nominal terms, but are more about buying mindshare than fiat. Small buys lead to big buys.

Properly sophisticated, grown-up, Ivy League MBA, *CFA-accredited* readers might liken this entire line of reasoning, and my enthusiasm for it, to GameStop, the finance hilarity du jour, in the sense of uppity retail thinking they are sticking it to the man but really just blowing their savings on a practical joke from which Citadel will be the ultimate winner (not my reading, to be clear, but a common one. Mine is that these people knew exactly what they were doing and that you can prove it if you are willing to just *look*). I would encourage such readers to think more seriously about the game theory involved in all of this, particularly if gold and then short-term *government* credit fall into Bitcoin's orbit.

You might think this leads precisely nowhere at all, but the central banks of Venezuela, Iran, North Korea, and Singapore would disagree with you, that we know of so far. Central bank accumulation will become the defining macroeconomic issue of the decade, and advocacy for accumulation by the tech-savvy one of the defining political issues. Entirely mobile, unseizable capital will be attracted to, and will compound physically, wherever it is most welcomed, as will the human capital that likely comes with it. Countries with geopolitical rivals who decide to ban Bitcoin will be cutting off their nose to spite their face. When China starts to pay Russia for natural gas first in dollar stablecoins on Bitcoin, then *in bitcoin*, don't say I didn't tell you to think about it a little more than not at all.

Come to think of it, even GameStop can be non-ironically tied to this discussion. It has turned out a decent amount of faux-populist rancor was misplaced and the real culprit for screwing the little guy wasn't shady backroom deals between Citadel, Sequoia, the SEC, and the Fed, but rather the limitations of equity clearing and settlement given the mechanics of counterparty risk. Try to imagine, if only for a moment, tokenized equity certificates pegged to a secure digital bearer asset, with no counterparties, that settle in T+*right now*. Can you imagine that? Citadel might be the short-term winner in all this, but the long term is all about tokenized equity on sidechains.

Bitcoin is Logos



photo by **Aaron Burden**, via Unsplash

> *"To imagine a language is to imagine a form of life."*
>
> *- Ludwig Wittgenstein*

LinkedIn founder Reid Hoffman infamously said on the Tim Ferris podcast that Bitcoin is like a Wittgenstein language game, with no elaboration whatsoever! Were I to pick up the baton, I would refer to *Philosophical Investigations* and the dictum that, "*the meaning of a word is its use in the language*." In other words (no pun intended), Hoffman is characterizing Bitcoin as understandable only by interpreting the actions of the participants as essentially consisting of communication with other

participants, hence expressing in a codified grammar what they *mean* and relying on this grammar to understand what others *mean.*

I think it is worth invoking Norwood Russell Hanson once again to appreciate the position of a functionally illiterate outsider in the presence of such a language game; an adherent of the (admittedly satirical) semantic theory of money. If perception is theory-laden, and if our theory invalidates the possibility of a new money monetizing from scratch, and that money takes the form of a language we don't speak, and we won't learn this language because we think it can't exist … we are just about *guaranteed* not to understand it.

That Bitcoin can be thought of as a language game clarifies why it is inherently peaceful. Money is an information system that records and updates who has done work that is valued by others, such that credit can be universalized and socially scaled. Frederic Lane and Reinhold Mueller note in *Money and Banking in Medieval and Renaissance Venice* that, "*both 'medium of exchange' and 'standard of value' are sufficiently ambiguous to make 'moneyness' a matter of degree,*" and that, "*conceptually and historically the two are separable.*"

This recording and updating is a technical problem, and candidate technical solutions ought to be evaluated not on how and how well they fit *sufficiently ambiguous* definitions, but on *how and how well they work;* be they Rai stones, gold ducats, or dollar-denominated bank liabilities; physical or digital; abstract or instantiated; debt or pure asset. The solution provided by Bitcoin is in some sense the purest yet conceived in that it captures this information *as speech* — we only use software to check the grammar.

As was made clear by Ross Stevens at SaylorCon, Ammous' framing of "salability across space" and "salability across time" is now firmly in the lexicon. It is worth teasing out further in the context of understanding Bitcoin as language, and money as an information system. The essence of temporal salability is soundnesss; the essence of spatial salability is portability. Prior to Bitcoin, the two were in inescapable tension; economic development induces a market demand for money to be increasingly purely informational, as commerce itself becomes more complex than movement of specie can efficiently support. But information is, by its nature, not scarce at all, and so retaining some semblance of scarcity in informational money, hence temporal salability, requires trust in a centralized source of truth.

Note this does not imply "fiat", but rather "fiduciary", from the Latin *fiducia*, for *trust*. With *relatively* sound specie held in reserve by *relatively* prudent bankers, "bank money" — payment purely by debit and credit of accounts with a bank—in thirteenth-century Venice was *relatively* trustworthy, and in fact dominant and thriving. Not to mention in Genoa, Florence, Barcelona, and Bruges, interoperable via bills of exchange; all a little less temporally salable, perhaps, but vastly more spatially salable. But of course, Bitcoin resolves the underlying tension entirely. It is digital (that is, *informational*) scarcity. We get the portability of email with no trust, just verification.



this historical and technical argument expressed in meme form

A Bitcoin transaction is a global speech act that means, roughly, *I am provably entitled to this portion, x, of the money supply, and am now transferring it to somebody else*, in a language that everybody remembers forever and which can't be used to lie. This is why efforts to ban Bitcoin, while they will certainly be attempted, will also almost certainly fail. Bitcoin is the ultimate samizdat. Bitcoin is logos.

This is also a source of great optimism given prevalent legal and social commitments to protecting politically undesirable speech. Naïve as I am sure it will seem to some, I think one of the most important US Supreme Court cases of the next 20 years will be the ruling that the right to broadcast Bitcoin transactions is guaranteed under the First Amendment. Prior to this,

while the legality is still up in the air, I fully expect a sitting congressperson to invoke congressional privilege and “broadcast” a transaction by dictating its hexadecimal representation on the floor of the House or Senate.

This will probably be followed by tweeting one, working one into a public deposition, embedding one in a flag — which will end up on t-shirts and lapel pins, as well as fully physically to be waved around as a staple at protests. *Don’t Tread on My Node!* You either make countably infinite sets of numbers, letters, and colors illegal — whatever that even means — or you accept that Bitcoin is going to happen.

Bitcoin is Techne

photo by **Garett Mizunaka**, via Unsplash

> *"The Venetians were not thinkers: they were doers. Empiricists par excellence, they mistrusted abstract theories."*
>
> *- John Julius Norwich*

Recording and updating the speech act of transferring value is a technical problem, to which Bitcoin is a technical solution. It is not an idea about how things *ought to work*. It is a real thing that *does work*. Although the observation may seem flippant, the distinction is enormously important.

Nic Carter put it to Frances Coppola recently that in order to make an apple pie from scratch, you have to first invent the universe, by which he was conveying that if you want to create a robust, fast-settling, online payments system with finality, you need to create Bitcoin. No alternative has ever *actually worked*. Given the problems Bitcoin solves are very clearly *not just academic* but are core to human civilization, *actually working* is rather important. Bitcoin is techne.

I think this captures what is likely the biggest hurdle for most newcomers who do actually make an effort to understand the details, because, on the face of it, Bitcoin is completely absurd as an engineering construct. *Miners do WHAT?!? Coins are stored HOW?!?* etc. — we've all had these conversations. Even the mathematically inclined who enjoy toying with the cryptographic primitives may very reasonably think, should they lack the wider context, *this is completely ridiculous because everything is fine*. But with the proper context, we can of course say, *this is exactly as ridiculous as it needs to be because everything is* ***not*** *fine.*

And yet there is a particular brand of skeptic who professes to admire some or most of Bitcoin's design, but can't bring themselves to get fully on board

because of some pet problem Bitcoin seems not to fully solve, or some pet issue with the way in which Bitcoin solves the problems it clearly does solve. Coppola is very much in this camp. I would argue Peter Schiff and Mike Green are too. All are interesting and serious people who seem to have the right attitude on many issues Bitcoin touches, with the mysterious exception *of Bitcoin itself*. On Bitcoin, all adopt variations of this pedantic quibbling that is superficially sophisticated but really the most meretricious and unserious position of all because nothing remotely practical is offered instead. They deal with reality not as it is but as they would like it to be. Their "solutions" are clean, slick, and are never going to happen. Bitcoin deals with reality as it actually is. It is ugly. And it works.

This entire line of argument will likely never go away, given it was refuted in its entirety as early as May 2011 in the now legendary *Bitcoin is Worse is Better*, by Gwern:

"*The sacrifice Bitcoin makes to achieve decentralization is — however practical — a profoundly* ugly *one. Early reactions to Bitcoin by even friendly cryptographers & digital currency enthusiasts were almost uniformly extremely negative, and emphasized the (perceived) inefficiency & (relative to most cryptography) weak security guarantees. Critics let 'perfect be the enemy of better' and did not perceive Bitcoin's potential. However, in an example of 'Worse is Better', the ugly inefficient prototype of Bitcoin successfully created a secure decentralized digital currency, which can wait indefinitely for success, and this was enough to eventually lead to adoption, improvement, and growth into a secure global digital currency*."

Embracing this engineering ethic will immunize the curious to such inane posturetalk as Nassim Taleb's unpredictable outbursts on "complex systems", "volatility", or "scale transformations," or Eric Weinstein claiming that, "*we need to get rid of the blockchain so that it's a locally enforced conservation law that*

replaces space-time with a system of computer nodes." If you want to *embed Bitcoin in a gauge theory*, Eric, go right ahead. I look forward to reading the BIP. But please *actually do it*. Don't burble Chomsky sentences in the metalanguage and expect to be taken seriously. Of colorless green ideas a decentralized currency is not made.

Bitcoin is Venice



> *"Of the various centres in which republican ideas continued to be discussed and celebrated throughout the later Renaissance, the one with the most enduring commitment to the traditional values of independence and self-government was Venice. While the rest of Italy succumbed to the rule of the signori, the Venetians never relinquished their traditional liberties."*

- Quentin Skinner

I tend to find Bitcoin analogies that aren't transparently rhetorical to inevitably have some fatal flaw that ultimately makes them more confusing than they are helpful. And yet, Venice has an enigmatic appeal that I cannot bring myself to categorize as entirely fanciful. As a social and political order emerging from feudalism by an embrace of trade and capital formation, it is certainly instructive. But there seems to me to be more. Clearly, Bitcoin is not a *city*, but it is a system, and a symbol, in a way that transcends its instantiation as code, much as Venice transcended its islands and lagoon.

Some comparisons are cute and easy. Venice was far, far easier to defend than to attack, to the point that attack was essentially futile. Its governance model was bewilderingly opaque and constitutionally resistant to seizure. If seizure nonetheless became a realistic threat, an immune response seemed to be triggered that innovated around the danger. Was the legendary putting down of Bajamonte Tiepolo's insurrection, tipped not by the Commune's security forces but by an old woman throwing a stone from a window, a primitive *user activated soft fork*? Sure, why not. And of course, what of emerging from a dark age characterized first and foremost by monetary debasement? Pirenne's observation certainly suggests a precedent,

"If it is admitted, as it must be admitted, that the reappearance of gold coinage, with the florins of Florence and the ducats of Venice in the thirteenth century characterized the economic renaissance of Europe, the inverse is also true: the abandoning of gold coinage in the eighth century was the manifestation of a profound decline."

What of Venice's relative egalitarianism? John Julius Norwich writes in *A History of Venice* that Venice was famed, "*for a system of justice which gave impartial protection to rich and poor, aristocrat and artisan, Venetian and foreigner; for, in theory, at any rate and for the most part in practice too, every*

man living beneath the banner of St Mark was equal in the sight of the law." Pirenne notes this attitude was fundamentally rooted in the necessities of commerce and hence extended beyond the city's own jurisdiction and internal affairs: "*No scruple had any weight with the Venetians. Their religion was a religion of businessmen. It mattered little to them that the Moslems were the enemies of Christ, if business with them was profitable*." Similarly, that Bitcoin is apolitical or "money for enemies" is well memeified at this point, but I was particularly struck by Terry Crewes' rather more visceral anecdote to this effect:

What of Venice's continual flouting of the proclamations of the Church, the literal counterpart to Kotkin's modern *clerisy* of elite tastemakers and thinkers of right thoughts, who already have and no doubt will continue to take, "*a defiant and hostile attitude toward the commercial revival which must, from the very first, have seemed to it a thing of shame and a cause of anxiety,*" as Pirenne put it?

But I think the most striking comparison of all is the synthesis of disparate ideas into a financial cornerstone. Very little of commercial note was invented in medieval and Renaissance Venice — double-entry bookkeeping was likely borrowed from Genoa, having originally been imported to Italy from the Levant; the numeral system with which it is most useful is famously Indian, relayed in Arabic via Persia, the Levant, and the Maghreb; most other contributions to business administration were probably imported from Arabia and Constantinople; and the material industrial advances of the time predominantly originated in China. But Venice combined them all to perfection. Most of the outlines of modern finance were arguably present in

Venice by the early fifteenth century at the latest, with very little truly *invented* since, rather than further combined, standardized, scaled, or modernized. To my mind, only central banking and options have been both material and entirely novel.

Of course, technology, industry, and society have advanced immeasurably since, and yet we still live by Venetian financial customs and have no idea why. Even the word "bank," in the financial sense, originates in Venice, from the *banca* or "benches" of moneychangers by the Rialto Bridge, with Lane and Mueller pointing out that, "*true banking had developed, it is now generally agreed, not from moneylending or pawnbroking, but from the manual exchange of coins.*" Modern banking is the legacy of a problem that technology has since solved.

The following explication of Venetian financial infrastructure around the fourteenth century from Lane's *Venice, A Maritime Republic* is remarkable in that I think it is a perfectly solid foundation for understanding the role played, *today*, by credit card networks and bank settlement schemes alike:

"*The main function of a Venetian banker was not making loans but making payments on behalf of his clients. Even if a merchant had plenty of coins in his treasure chest, it was a bothersome and dangerous business to get them out every time he made a purchase, making sure each coin was genuine and in good condition. Nor did he want to go through a similar process each time he made a sale. He was happy to receive payment by being given credit on the books of a well-known banker. He could use that credit to pay for his next purchase. These credits were not transferred through writing checks, as is done today, but depended on the person who was making a payment appearing in person before the banker who sat behind a bench under the portico of a church at Rialto, with his big journal spread out in front of him. The payor orally instructed the banker to make a transfer to*

the account of the person being paid. The banker wrote as directed in his book, which was an official notarial record, so that there was no need of receipts. There were normally four or five such bankers with booths on the campo next to the Rialto bridge. Everyone of any consequence in business had an account so that he could make and receive payments through the banks. They were called banche di scritta *or* del giro *because their main function was to write transfers and thus to rotate (*girare*) credits from one account to another at the command of the merchants.*"

If we add bills of exchange, credit creation, and floating the state debt, all natural extensions of the utility of these *ledgers* — and of course if we subtract a hard reserve asset, available on-demand, that was deposited in the first place — there is not much left to account for. And note as well the seedlings of why the trust-minimizing natively digital, computational, and decentralized features of Bitcoin for settlement and Lightning for payments dramatically improve on this setup. Bitcoin may be magic Internet money, but more importantly, it is money *for the Internet.* Prior to 2009, you could send any information you want to anybody, anywhere in the world, instantly … *except* the most important information of all: value. Now we are all caught up.

It is often commented that Bitcoin is really more an ingenious combination of prior advances in applied cryptography than an *invention* in its own right. I am quite partial to the romantic idea that Bitcoin was discovered rather than invented. It is a foundation to scale the next great phase of economic progress. Bitcoin is Venice.

Bitcoin Is



photo by **Christian Puta**, via Unsplash

> *"Our history forbids us to be surprised that an orthodoxy of thought should become narrow, rigid, mercenary, morally corrupt, and vengeful against dissenters. This has happened over and over again. It might be thought the maturity of orthodoxy; it is what finally happens to a mind once it has consented to be orthodox. But one may be permitted a little amusement, if not surprise, that this should have befallen a modern science, which was set up, as it never tires of advertising, to pursue truth, not protect it … If change is to come, then, it will have to come from outside. It will have to come from the margins."*
>
> *- Wendell Berry*

Colorfully outrageous metaphors finally aside, the most remarkable fact of all remains that Bitcoin even exists. Bitcoin *is*. This is undeniable, although the reasons why can be ignored — our monetary system is optimized to strip-

mine capital — and its ascent can be misunderstood — the mainstream understanding of money invalidates it by definition rather than observation.

Ludwig Wittgenstein once asked a friend, "*tell me, why do people say it is more natural to think that the sun rotates around the earth than that the earth is rotating?"* The friend said, "*well, obviously, because it just* ***seems like*** *the sun is going around the earth.*" Wittgenstein replied, "*well, what would it* ***seem like*** *if it did* ***seem like*** *the earth were rotating?*"

If it seemed like a global, digital, sound, open source, programmable money was monetizing from absolute zero, it would seem a lot like this.

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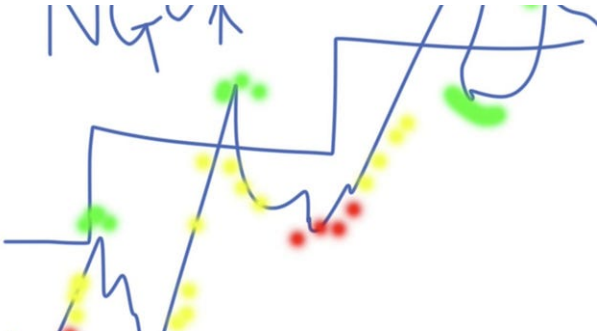
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